```
import gymnasium as gym
import numpy as np


def run_random_agent(env_name: str, num_episodes: int = 100):
    """Executes a Random Agent in the specified Gymnasium environment.

    Calculates and displays total and average rewards across episodes.
    """
    # 1. Create the environment
    env = gym.make(env_name)

    episode_rewards = []

    print(f"\n{'='*50}")
    print(f" Running Random Agent on: {env_name}")
    print(f" Action Space: {env.action_space}")
    print(f" Observation Space: {env.observation_space}")
    print(f"{'='*50}")

    for episode in range(1, num_episodes + 1):
        # 2. Reset environment and 3. Observe initial state
        state, info = env.reset()

        terminated = False
        truncated = False
        total_reward = 0.0

        while not (terminated or truncated):
            # 4. & 5. Identify action space and select a random action
            action = env.action_space.sample()

            # 6. Execute action, 7. Observe next state, reward, and terminal signals
            next_state, reward, terminated, truncated, info = env.step(action)

            total_reward += reward
            state = next_state

        episode_rewards.append(total_reward)

        if episode <= 5 or episode % 25 == 0:
            print(
                f"Episode {episode:3d}/{num_episodes} | Total Reward:"
                f" {total_reward:.2f}"
            )

    # 9. Calculate total and average metrics across all episodes
    overall_total_reward = sum(episode_rewards)
    average_reward = np.mean(episode_rewards)

    print(f"\n--- Summary for {env_name} ---")
    print(
        f"Total Cumulative Reward ({num_episodes} episodes):"
        f" {overall_total_reward:.2f}"
    )
    print(f"Average Reward per Episode: {average_reward:.2f}")

    env.close()
    return episode_rewards


if __name__ == "__main__":
    # Required target environments
    environments = ["CartPole-v1", "MountainCar-v0", "FrozenLake-v1"]

    for env_id in environments:
        run_random_agent(env_id, num_episodes=100)
```

```
==================================================
 Running Random Agent on: CartPole-v1
 Action Space: Discrete(2)
 Observation Space: Box([-4.8               -inf -0.41887903        -inf], [4.8               inf 0.41887903        inf],
==================================================
Episode   1/100 | Total Reward: 13.00
Episode   2/100 | Total Reward: 16.00
Episode   3/100 | Total Reward: 16.00
Episode   4/100 | Total Reward: 12.00
Episode   5/100 | Total Reward: 12.00
Episode  25/100 | Total Reward: 19.00
Episode  50/100 | Total Reward: 28.00
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Episode  75/100 | Total Reward: 33.00
Episode 100/100 | Total Reward: 14.00

--- Summary for CartPole-v1 ---
Total Cumulative Reward (100 episodes): 2163.00
Average Reward per Episode: 21.63

==================================================
 Running Random Agent on: MountainCar-v0
 Action Space: Discrete(3)
 Observation Space: Box([-1.2  -0.07], [0.6  0.07], (2,), float32)
==================================================
Episode   1/100 | Total Reward: -200.00
Episode   2/100 | Total Reward: -200.00
Episode   3/100 | Total Reward: -200.00
Episode   4/100 | Total Reward: -200.00
Episode   5/100 | Total Reward: -200.00
Episode  25/100 | Total Reward: -200.00
Episode  50/100 | Total Reward: -200.00
Episode  75/100 | Total Reward: -200.00
Episode 100/100 | Total Reward: -200.00

--- Summary for MountainCar-v0 ---
Total Cumulative Reward (100 episodes): -20000.00
Average Reward per Episode: -200.00

==================================================
 Running Random Agent on: FrozenLake-v1
 Action Space: Discrete(4)
 Observation Space: Discrete(16)
==================================================
Episode   1/100 | Total Reward: 0.00
Episode   2/100 | Total Reward: 0.00
Episode   3/100 | Total Reward: 0.00
Episode   4/100 | Total Reward: 0.00
Episode   5/100 | Total Reward: 0.00
Episode  25/100 | Total Reward: 0.00
Episode  50/100 | Total Reward: 0.00
Episode  75/100 | Total Reward: 0.00
Episode 100/100 | Total Reward: 0.00

--- Summary for FrozenLake-v1 ---
Total Cumulative Reward (100 episodes): 1.00
Average Reward per Episode: 0.01
```